FINAL TERM

Physics 1 [Summer 2024-2025]

Department of Physics
Faculty of Science & Technology (FST)
American International University-Bangladesh

Reference Books

- ***Fundamentals of Physics (Edition: 10th) Written by Halliday, Resnick and Walker***

- ***University Physics (13/14th Edition) written by Hugh D. Young and Roger A. Freedman***

LESSON 1

BOOK CHAPTER 22

ELECTRIC FIELDS

Think, why……..



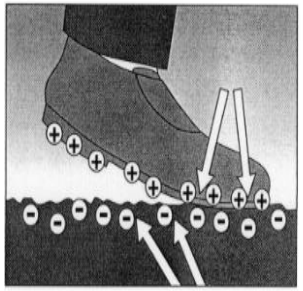
Taking off the hat in the winter

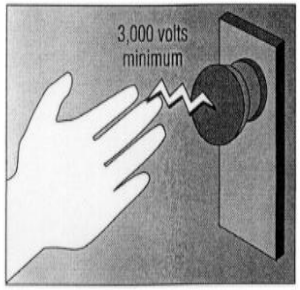


Attraction between the balloon and hair

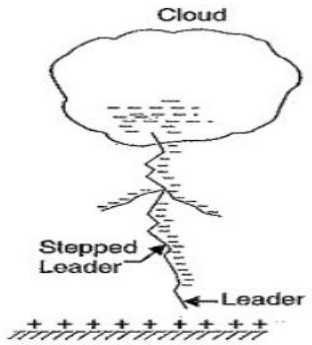


Lightning





Static charge is responsible for all of these.

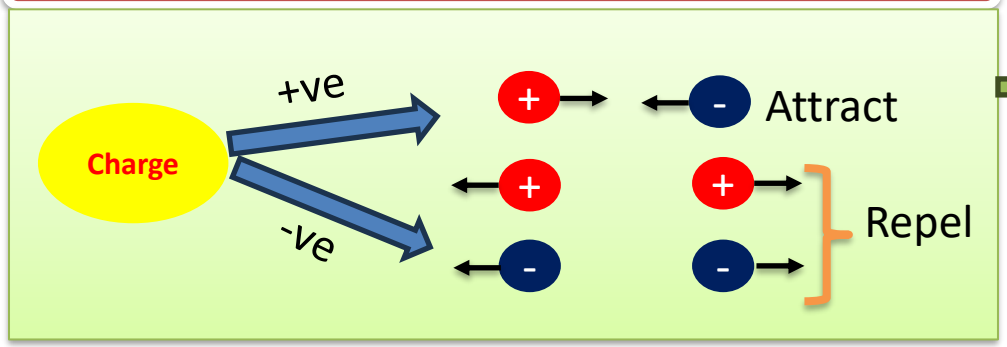


Why do you feel the shock when you scuff your shoes across a carpet and then reach for a metal doorknob?

This is due to charged particles leaping between your finger and the doorknob.

Nature of electric charge

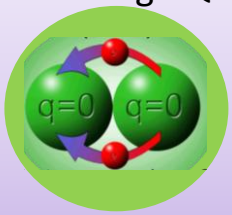
Like charges repel, unlike charges attract



- **A positive charge and a negative charge attract each other.**
- **Two positive charges or two negative charges repel each other.**

Charge is a conserved quantity.

Net charge Q = 0



gains 2 electrons loses 2 electrons

The principle of conservation of charge:
The algebraic sum of all the electric charges in any closed system is constant.

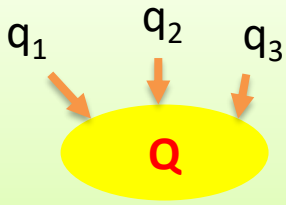
Electric Charge is quantized.

Q

Q = n x e

The magnitude of charge of the electron or proton is a natural unit of charge.
Every observable amount of electric charge is always an integer multiple of this basic unit. That is charge is *quantized*

Charges are additive in nature.



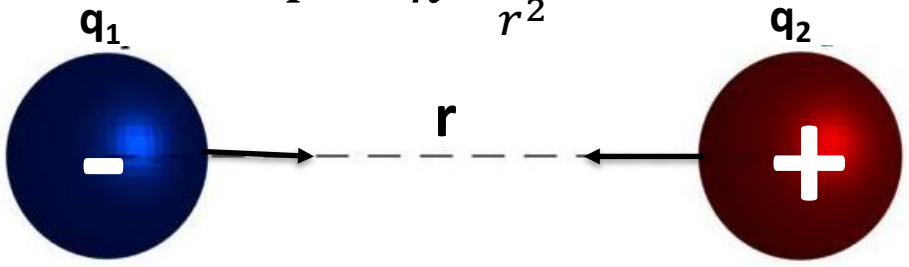
$Q = q_1 + q_2 + q_3$

The charge of an electron or a proton,
$e = 1.6021764871402 \times 10^{-19}$ C

Coulomb's Force (Electrostatic force)

Coulomb's law: (Charles Augustin de Coulomb)
The magnitude of the electric force between two-point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$F = k\frac{q_1 q_2}{r^2}$$



The magnitude **F** of the force that each of two-point charges q_1 and q_2 , a distance r apart, exerts on other can be expressed as

$$\boldsymbol{F = k\frac{q_1 q_2}{r^2}}$$

$$or, F = \frac{1}{4\pi\epsilon_o}\frac{q_1 q_2}{r^2}$$

$$\frac{1}{4\pi\epsilon_0} = k = 8.988 \times 10^9\ \mathrm{N \cdot m^2/C^2}$$

$$\epsilon_0 = 8.854 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}$$

$$e = 1.6021764871402 \times 10^{-19}\ \mathrm{C}$$

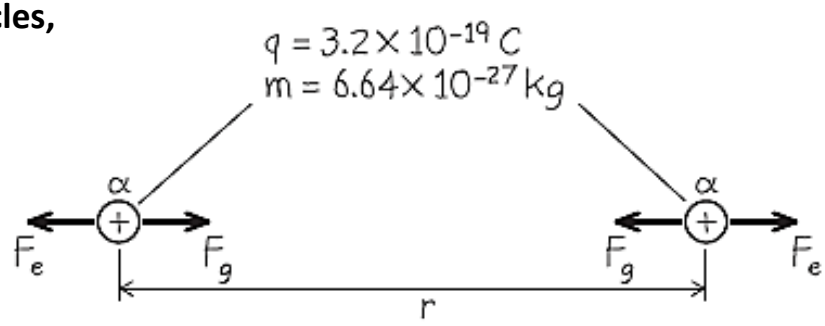
The directions of the forces the two charges exert on each other are always along the line joining them.

Electric force versus gravitational force

An α particle (the nucleus of a helium atom) has mass m = 6.64 x 10^{-27} kg and charge q = 3.2 x 10^{-19} C. Compare the magnitude of the electric repulsion between two α ("alpha") particles with that of the gravitational attraction between them.

Answer: **The Electric repulsion force between two α particles,**

$$F_e = \frac{1}{4\pi\epsilon_o}\frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_o}\frac{q^2}{r^2}$$



The gravitational attractive force between two α particles,

$$F_G = G\frac{m_1 m_2}{r^2} = G\frac{m^2}{r^2}$$

The ratio of electric force to gravitational force,

$$\frac{F_e}{F_G} = \frac{1}{4\pi\epsilon_o G}\frac{q^2}{m^2} = \frac{9\times10^9 N.m^2/C^2}{6.67\times10^{-11}N.m^2/Kg^2}\frac{(3.2\times10^{-19}C)^2}{(6.64\times10^{-27}Kg)^2}$$

$$= \mathbf{3.1\times10^{35}}$$

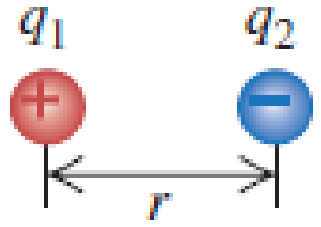
This astonishingly large number shows that the gravitational force in this situation is completely negligible in comparison to the electric force.

Gravitational force always attractive but electrostatic force may be attractive or repulsive depends on charge types.

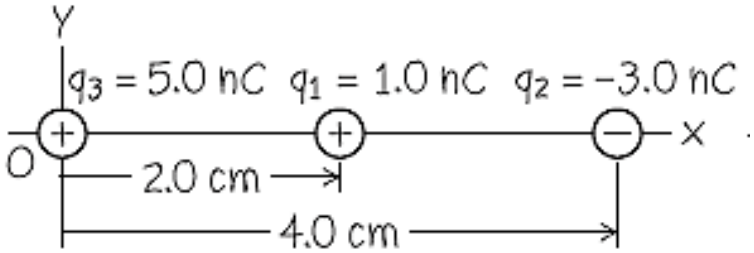
Force between two-point charges

Solve it by yourself !

1. Two point charges, q_1= 25 nC and q_2 = -75 nC are separated by a distance r = 3 cm (Fig. a). Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1



2. Two point charges are located on the x-axis of a coordinate system: q_1= 1 nC is at x = +2.0 cm and q_2 = -3.0 nC is at x = +4 cm . What is the total electric force exerted by q_1 and q_2 and on a charge q_3 = 5.0 nC at x = 0?



ELECTRIC FIELD AND ELECTRIC FORCES

Think about it.....

When two electrically charged particles in empty space interact, How does particle 1 "know" of the presence of particle 2? That is, since the particles do not touch, how can particle 2 push on particle 1—how can there be such an *action at a distance*?

We can answer those questions by saying that particle 2 sets up an ***electric field*** in the space surrounding itself. If we place particle 1 at any given point in that space, the particle "knows" of the presence of particle 2 because it is affected by the electric field that particle 2 has already set up at that point. Thus, particle 2 pushes on particle 1 not by touching it but by means of the electric field produced by particle 2.

Can you recall that Gravity can also be thought of as an "action-at-a-distance" force or noncontact force?

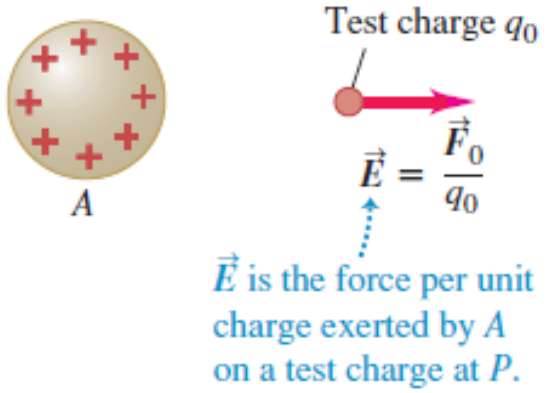
ELECTRIC FIELD

An electric field is a vector field because it is responsible for conveying the information for a force, which involves both magnitude and direction. This field consists of a distribution of electric field vectors $\vec{E}$, one for each point in the space around a charged object.

We define the electric field $\vec{E}$ at a point as the electric force $\vec{F}$ experienced by a test charge q_0 at the point, divided by the charge q_0.That is, *the electric field at a certain point is equal to the electric force per unit charge experienced by a charge at that point:*

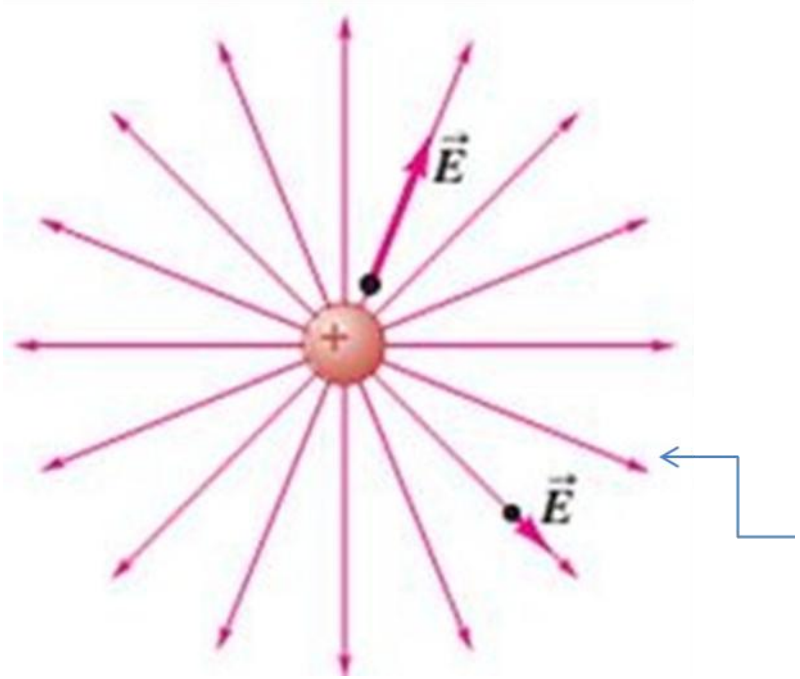
$$\vec{E} = \frac{\vec{F}}{q_0}$$

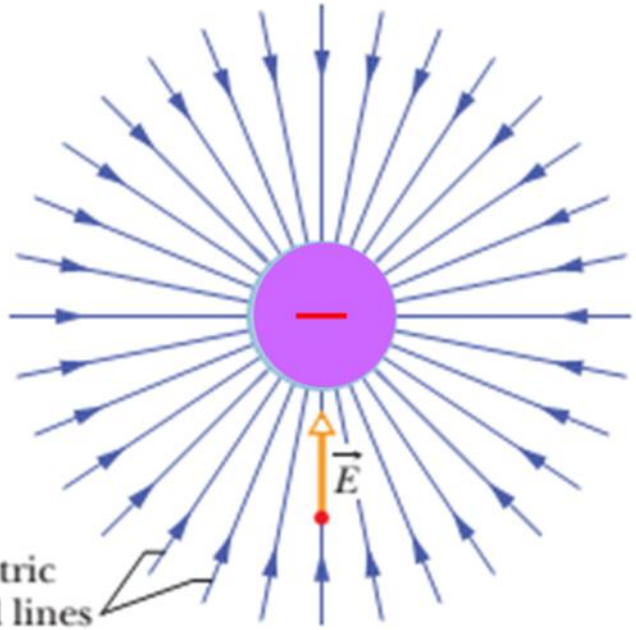
The SI unit for the electric field is the Newton per Coulomb (N/C).



- **A charged body creates an electric field in the space around it.**
- **The electric force on a charged body is exerted by the electric field created by *other* charged bodies.**

Electric Field lines around charged particles (Point Charges)

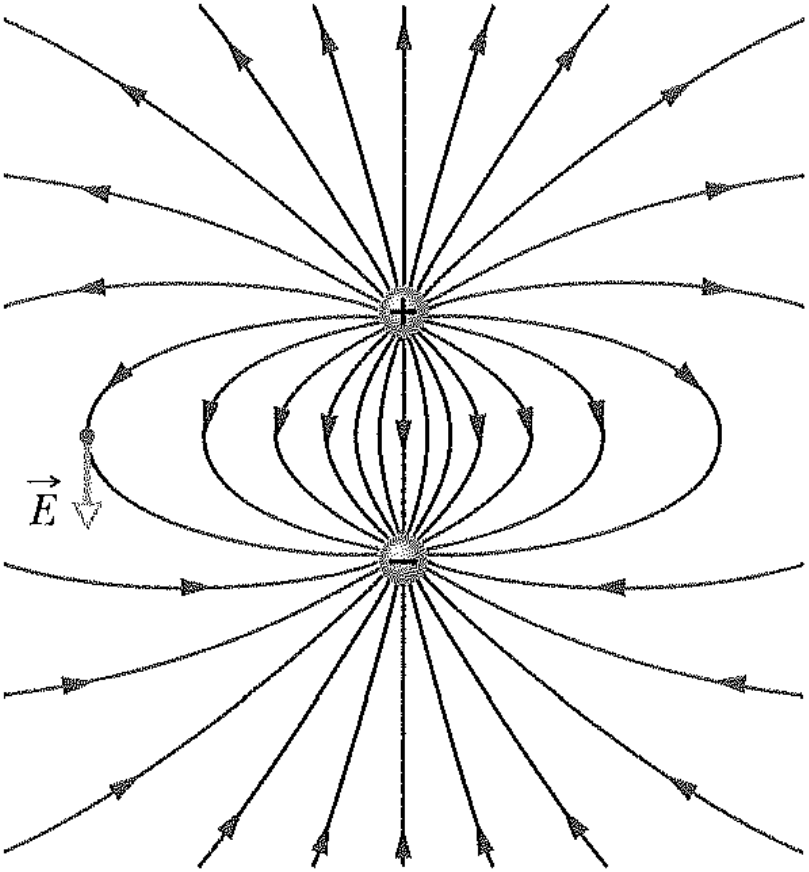




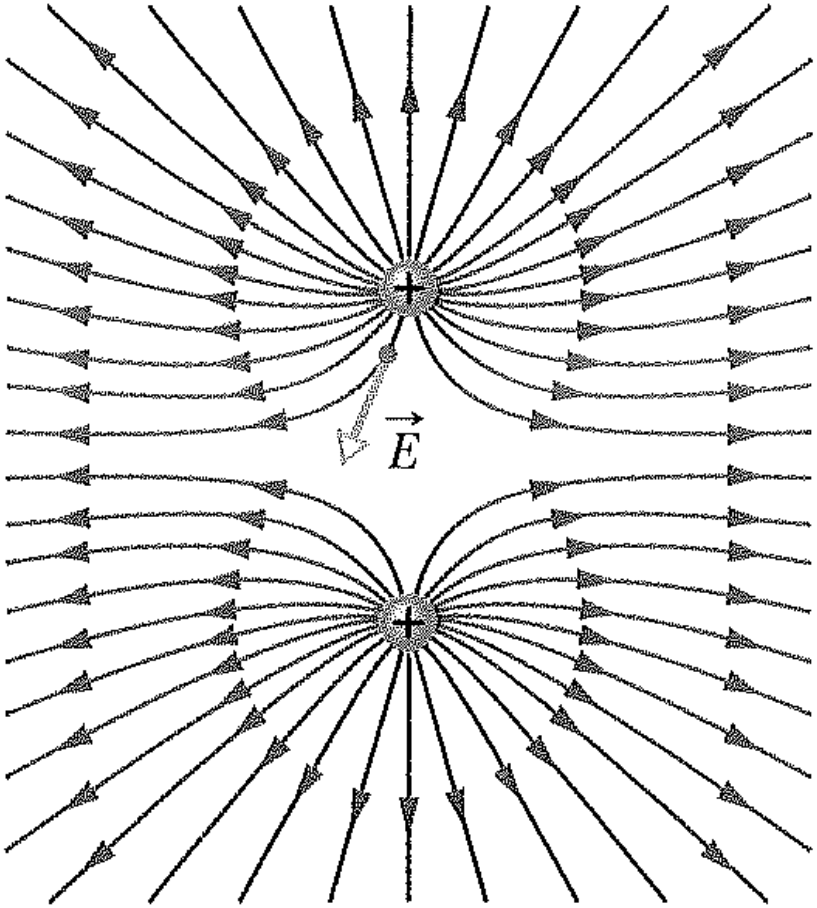
The field produced by a positive point charge points away from the charge

The field produced by a negative point charge points toward the charge

Electric field lines extend away from positive charge (where they originate) and toward negative charge (where they terminate).



Field lines for a positive point charge and a nearby negative point charge that are equal in magnitude.



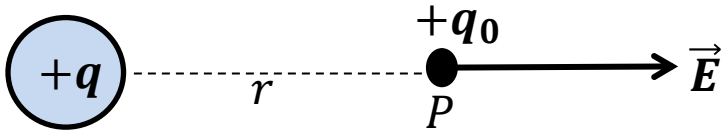
Field lines for two equal positive point charges.

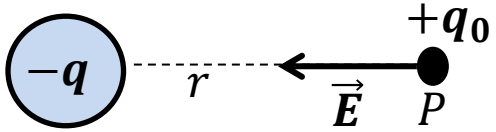
Electric field lines help us visualize the direction and magnitude of electric fields. ***The electric field vector at any point is tangent to the field line through that point. The density of field lines in that region is proportional to the magnitude of the electric field there. Thus, closer field lines represent a stronger field.***

Electric Field due to a Point Charge:

If we place a small test charge $+q_0$ at the field point P at a distance r from the point charge q, the magnitude F of the force is given by the Coulomb's law,

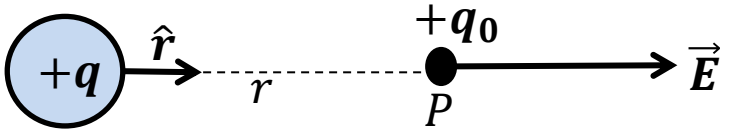
$$F = \frac{1}{4\pi\varepsilon_0}\frac{|qq_0|}{r^2}$$





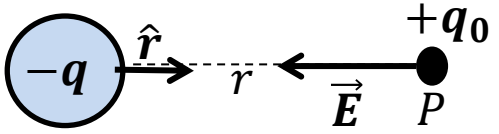
The quantity $\boldsymbol{\varepsilon_0}$, called the **permittivity constant. The value of $\boldsymbol{\varepsilon_0}$ is**

$$\varepsilon_0 = 8.85 \times 10^{-12}\ \frac{C^2}{N.m^2}$$



The magnitude E of the electric field at point P is

$$E = \frac{F}{q_0} = \frac{\frac{1}{4\pi\varepsilon_0}\frac{|qq_0|}{r^2}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$



$$\boxed{E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}}$$

In vector form,

$$\boxed{\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}}$$

Problem 5 (Book chapter 22)

A charged particle produces an electric field with a magnitude of 2.0 N/C at a point that is 50 cm away from the particle. What is the magnitude of the particle's charge?

Answer: Required formula:

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

$$2 = 9 \times 10^9 \frac{q}{(0.50)^2}$$

$$q = \frac{(2)(0.25)}{9 \times 10^9} = 0.0555 \times 10^{-9}\ C$$

Given

$E = 2.0\ N/C$

$r = 50\ cm = 0.50\ m$

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N.m^2/C^2$

$q = ?$

Problem 6 (Book chapter 22)

What is the magnitude of a point charge that would create an electric field of 1.00 N/C at points 1.00 m away?

Answer: Follow exactly similar steps as Problem 5 to answer problem 6.

Let's try !!!!

1. Calculate electric field at point C for figure 1.

2. Calculate net electric field at point C for fig. 2.

3. Calculate net electric field at the mid point of the distance between the two point charges for fig. 3.

Fig. 1

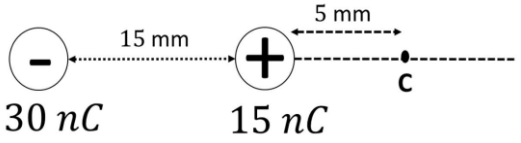


Fig. 2

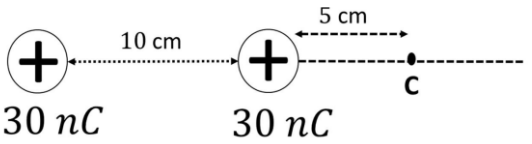
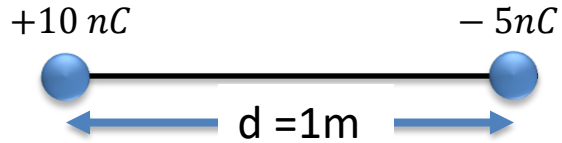
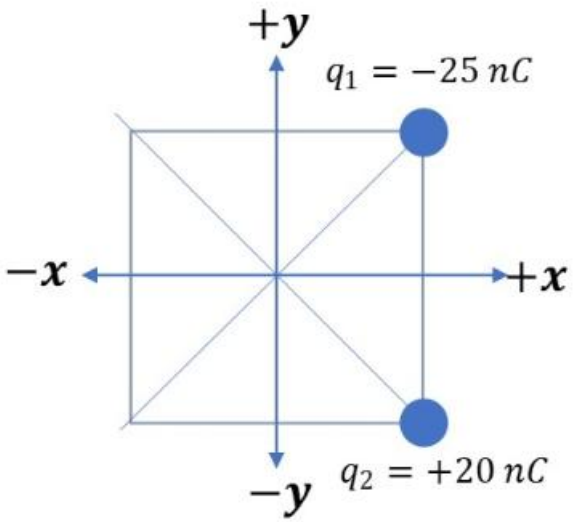


Fig. 3



4. In the adjacent figure, two point charges are at the corner of a square of edge length a = 7.00 cm . In unit-vector notation, what net electric field do the particles produce at the square's center?



12987.2 i + 116888.6 j

Electric Dipoles

An electric dipole is a pair of point charges with equal magnitude and opposite sign (a positive charge $+q$ and a negative charge $-q$) separated by a small distance d.

Fig.1

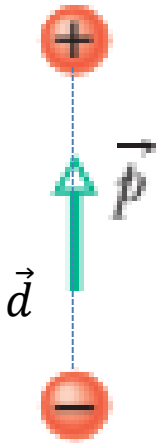
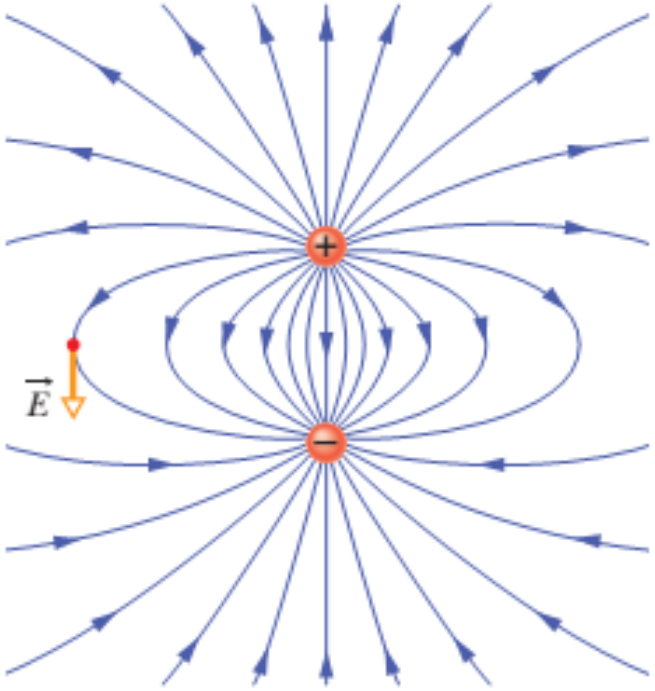


Fig.2



The product of the charge q and the separation d is the magnitude of a quantity called the **electric dipole moment,** denoted by p.

$$That\ is, \qquad p = qd$$

In vector form, $\vec{p} = q\vec{d}$

The direction of $\vec{p}$ is from negative charge to positive charge as shown in figure 1.

The pattern of electric field lines around an electric dipole, with an electric field vector $\vec{E}$ shown (Figure 2) at one point (tangent to the field line through that point).

The Electric Field Due to an Electric Dipole:

The net magnitude of the electric field at point P is

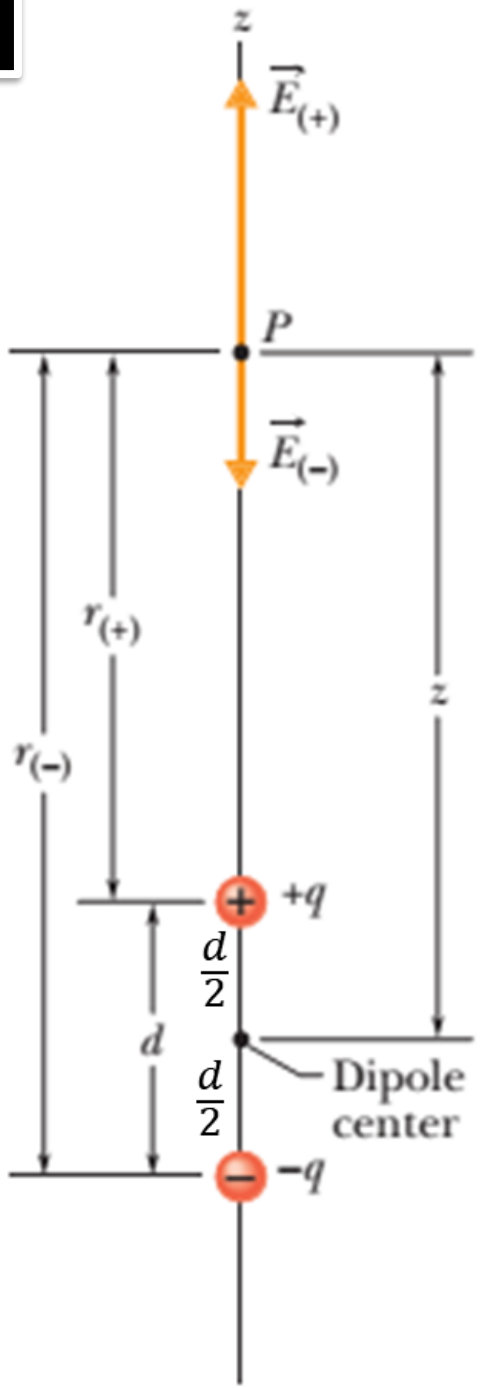
$$E = E_{(+)} - E_{(-)}$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r_{(+)}^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{r_{(-)}^2}$$

$$E = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\left(z-\frac{d}{2}\right)^2} - \frac{1}{\left(z+\frac{d}{2}\right)^2}\right]$$

$$E = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\left\{z\left(1-\frac{d}{2z}\right)\right\}^2} - \frac{1}{\left\{z\left(1+\frac{d}{2z}\right)\right\}^2}\right]$$

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\left[\frac{1}{\left(1-\frac{d}{2z}\right)^2} - \frac{1}{\left(1+\frac{d}{2z}\right)^2}\right]$$



$$E = \frac{q}{4\pi\varepsilon_0 z^2}\left[\left(1-\frac{d}{2z}\right)^{-2} - \left(1+\frac{d}{2z}\right)^{-2}\right]$$

For $z \gg d$, we have $\frac{d}{2z} \ll 1$

We use the form of binomial theorem,

$$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \ldots \quad (x^2 < 1)$$

$$\boldsymbol{(1+x)^n \approx 1 + nx \quad for\ x \ll 1}$$

$Here \quad \frac{d}{2z} \ll 1$

$Therefore, we\ can\ write,$

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\left[\left(1+\frac{2d}{2z}\right) - \left(1-\frac{2d}{2z}\right)\right]$$

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\left[1+\frac{d}{z}-1+\frac{d}{z}\right]$$

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\left[2\frac{d}{z}\right]$$

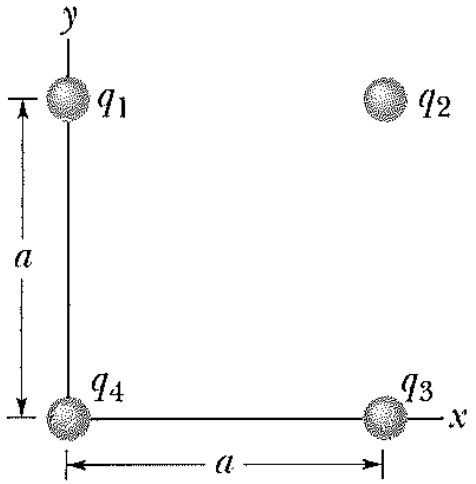
$Finally,$

$$E = \frac{2qd}{4\pi\varepsilon_0 z^3} = \frac{2p}{4\pi\varepsilon_0 z^3} = \frac{p}{2\pi\varepsilon_0 z^3}$$

$here\ p = qd = electric\ dipole\ moment$

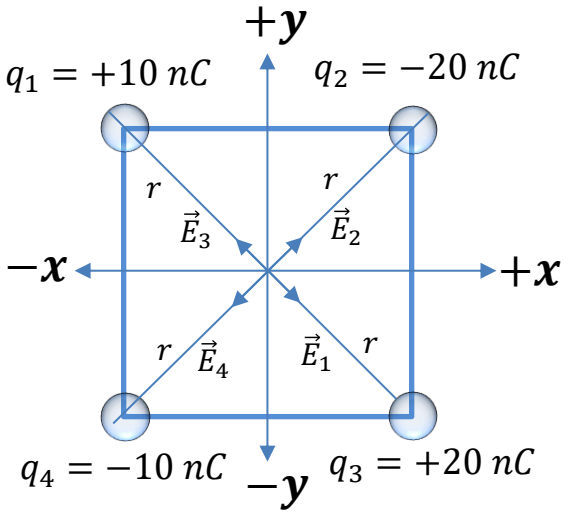
Problem 7 (Book chapter 22)

In the adjacent figure, the four particles form a square of edge length *a* = 5.00 cm and have charges $q_1 = +10\,nC$, $q_2 = -20\,nC$, $q_3 = +20\,nC$, and $q_4 = -10\,nC$. In unit-vector notation, what net electric field do the particles produce at the square's center?



Answer:

The net electric field at the center of the square along x-axis is



$$E_x = E_2 cos45^0 + E_1 cos45^0 - (E_3 cos45^0 + E_4 cos45^0)$$

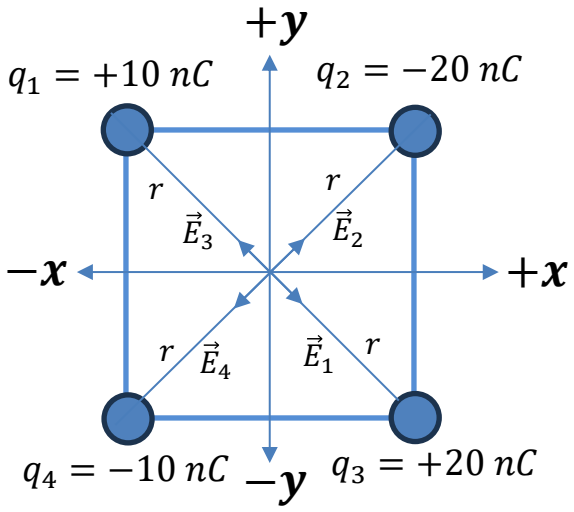
Here,

$$E_1 = E_4 = \frac{1}{4\pi\varepsilon_0}\frac{10\times10^{-9}}{r^2} \text{ And } \quad E_3 = E_2 = \frac{1}{4\pi\varepsilon_0}\frac{20\times10^{-9}}{r^2}$$

$$E_x = E_2 cos45^0 + E_1 cos45^0 - E_2 cos45^0 - E_1 cos45^0$$

E_x =0 N/C

The net electric field at the center of the square along y-axis is



$$E_y = E_2 cos45^0 + E_3 cos45^0 - (E_1 cos45^0 + E_4 cos45^0)$$

$$E_y = E_2 cos45^0 + E_2 cos45^0 - (E_1 cos45^0 + E_1 cos45^0)$$

$$E_y = 2E_2 cos45^0 - 2E_1 cos45^0$$

$$E_y = \frac{2 \times 9 \times 10^9 \times 20 \times 10^{-9} \times 0.707}{r^2} - \frac{2 \times 9 \times 10^9 \times 10 \times 10^{-9} \times 0.707}{r^2}$$

$$r = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \sqrt{\frac{2a^2}{4}}$$

$$r = \frac{a}{\sqrt{2}} = \frac{0.05}{1.414} = 0.0354\ m$$

$$E_y = \frac{127.26}{r^2} = \frac{127.26}{(0.0354)^2} = 101.55 \times 10^3\ N/C$$

$$\vec{E} = E_x \hat{\imath} + E_y \hat{\jmath} = 0 + 101.55 \times 10^3\ \hat{\jmath}$$

$$= 101.55 \times 10^3\ \hat{\jmath}\ \frac{N}{C}$$

Linear charge density:

When charge is distributed along a line (such as a long, thin, charged plastic rod), we use (the Greek letter lambda λ) to represent the charge per unit length known as **linear charge density.**

That is

$$\lambda = \frac{Amount\ of charge\ distributed\ on\ the\ rod}{Length\ of\ the\ rod}$$

[For uniform linear charge density]

The SI unit of λ is Coulomb/meter; simply, we use C/m.

Note:

Some Measures of Electric Charge

Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m^2
Volume charge density	ρ	C/m^3

THANK YOU